Convergence to spurious minima

Theorem

If $\lambda \leq \lambda^*$, the gradient flow converges for any $m \in \mathbb{N}$ to some θ_∞ such that

$$h_{\theta_\infty}(x) = x^\top \beta^* \text{ for any } x$$

- $L(\theta_\infty) = \Omega(1)$
- **λ^* does not depend on m**
- holds for a set of data of non-zero measure (non-degenerate)

Convergence in rich regime

Contradicts Chizat & Bach (2018)?

Theorem (informal) [Chizat & Bach '18]

For smooth activations and the initialisation $\begin{cases} w_i \sim \frac{\lambda}{\sqrt{m}} \mathcal{U}(\mathbb{S}_{d-1}) \\ a_i \sim \frac{\lambda}{\sqrt{m}} \mathcal{U}(\{-1,1\}) \end{cases}$

$$\lim_{m \to \infty} \lim_{t \to \infty} L(\theta_m^t) = 0$$